

Entity Relationship Diagram

1. Introduction

This document presents the Entity Relationship Diagram (ERD) and the accompanying Data Dictionary for the Laundry Shop Management System. Together, these artifacts define the system's logical data structure, specifying the entities, attributes, primary keys, foreign keys, relationships, and data constraints required to support the system's business processes.

The ERD and Data Dictionary are derived directly from the validated Data Flow Diagram (DFD) and approved functional requirements, ensuring consistency and traceability across system analysis and design outputs.

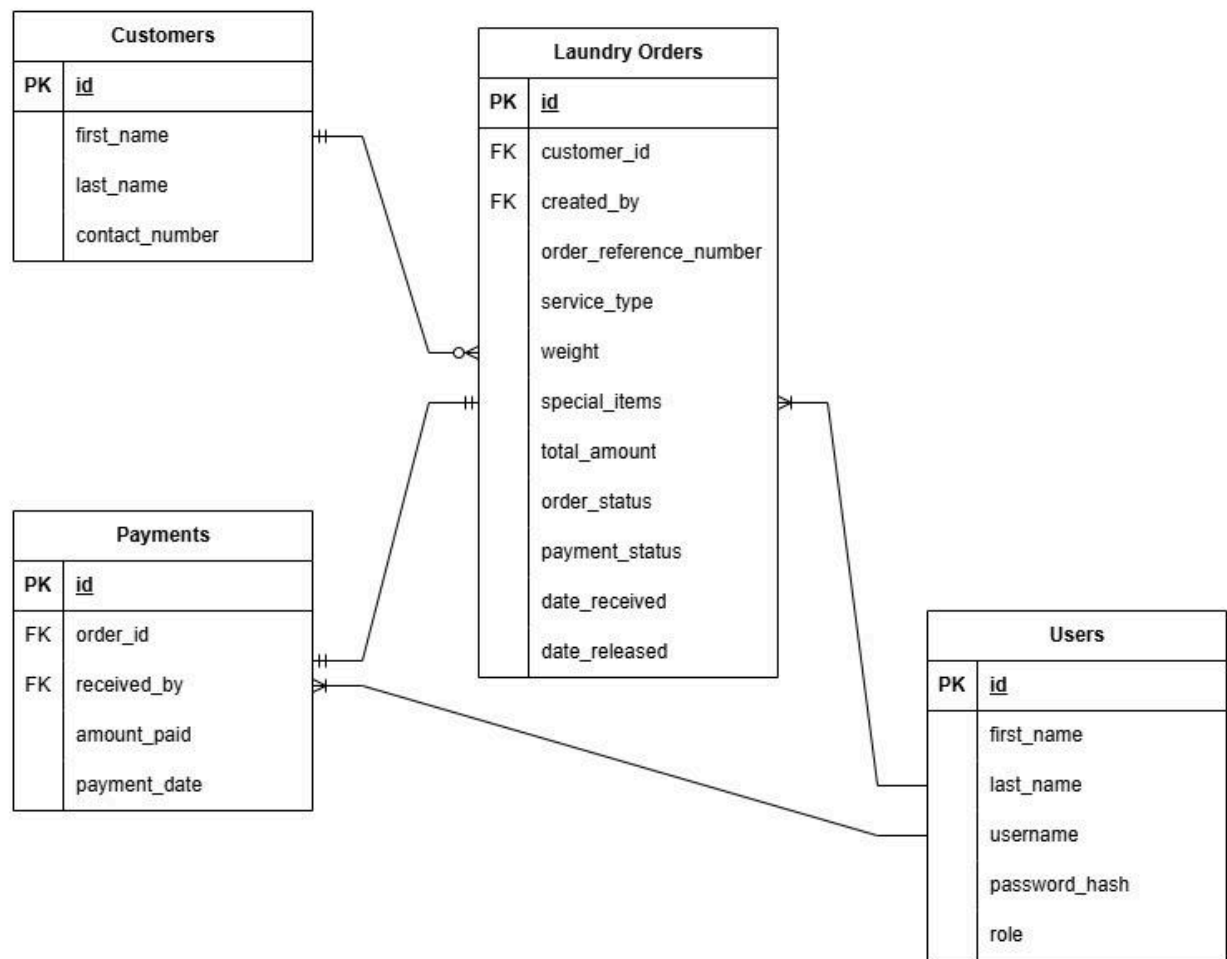
2. Purpose of the Document

The purpose of this document is to:

- Define the system's core data entities and relationships
- Provide a clear and unambiguous description of each data element
- Establish data integrity rules and constraints
- Serve as a foundation for database implementation
- Support later phases such as form design, reporting, and system architecture

3. Entity Relationship Diagram (ERD)

Laundry Shop Management System | Entity Relationship Diagram



4. Data Dictionary

4.1. Customer Table

Field Name	Descripti on	Data Type	Length	Required	Key / Constrain t	Example
customer_ id	Unique customer identifier	INT	–	Yes	PK	1001
first_name	Customer first name	VARCHA R	100	Yes	–	Juan
last_name	Customer last name	VARCHA R	100	Yes	–	Dela Cruz

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contact_number	Customer contact number	VARCHAR	15	Yes	–	09171234567
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4.2. User Table

Field Name	Description	Data Type	Length	Required	Key / Constraint	Example
user_id	Unique user identifier	INT	–	Yes	PK	1001
first_name	Customer first name	VARCHAR	100	Yes	–	Mary
last_name	Customer last name	VARCHAR	100	Yes	–	Dela Cruz
username	Login username	VARCHAR	50	Yes	Unique	staff01
password_hash	Encrypted password	VARCHAR	255	Yes	–	*****
role	System role	ENUM	–	Yes	Owner / Staff	Staff

4.3. LaundryOrder Table

Field Name	Description	Data Type	Length	Required	Key / Constraint	Example
order_id	Unique order identifier	INT	–	Yes	PK	5001
order_reference_number	Claim reference number	VARCHAR	20	Yes	Unique	ORD-2026-001
customer_id	Linked customer	INT	–	Yes	FK → Customer	1001

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created_by	User who created order	INT	–	Yes	FK → User	1002
service_type	Type of laundry service	ENUM	–	Yes	Wash/Dry/Fold	Wash
weight_kg	Laundry weight	DECIMAL	5,2	Yes	–	6.50
special_items	Special notes/items	VARCHAR	255	No	–	Comforter
total_amount	Total service cost	DECIMAL	8,2	Yes	–	350.00
order_status	Processing status	ENUM	–	Yes	Received/Ready	Ready
payment_status	Payment indicator	ENUM	–	Yes	Paid/Unpaid	Paid
date_received	Date order was created/received	DATE	–	Yes	–	2026-01-30
date_released	Date order was released	DATE	–	Yes	–	2026-01-30

4.4. Payment Table

Field Name	Description	Data Type	Length	Required	Key / Constraint	Example
payment_id	Unique payment identifier	INT	–	Yes	PK	9001

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order_id	Related laundry order	INT	–	Yes	FK → LaundryOrder (Unique)	5001
received_by	User who received payment	INT	–	Yes	FK → User	1002
payment_date	Date of payment	DATE	–	Yes	–	2026-01-30
amount_paid	Payment amount	DECIMAL	8,2	Yes	–	350.00